# Assignment#1

Q: Write a program that creates 2 child processes to calculate grades of students of mathematics class. The parent process will send the array to one child processes to calculate the percentage of marks out of 250 total. The other child processes will assign the grade A+>=90, A≥80, B≥70, C≥60, D≥50, F< and send result to parent process for sorting in ascending order and display the final result of the students. Class strength of 50 students

- Declare and initialize the array
- Each child process should run on a separate CPU
- Binding process to a CPU called affinity



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01 ⇒ run assignment-01
ID    | Name              | Marks    | Percentage   | Grade
-------------------------------------------------------------
11    | Usman Ghani       | 135.50   | 54.20%       | D
5     | Bilal Hassan      | 142.00   | 56.80%       | D
40    | Salman Khan       | 145.30   | 58.12%       | D
25    | Haris Rauf        | 148.60   | 59.44%       | D
30    | Kashif Ali        | 150.90   | 60.36%       | C
50    | Imran Khan        | 152.70   | 61.08%       | C
17    | Daniyal Shah      | 155.20   | 62.08%       | C
46    | Ali Haider        | 158.40   | 63.36%       | C
34    | Waqar Younis      | 159.00   | 63.60%       | C
13    | Saad Abdullah     | 160.50   | 64.20%       | C
28    | Shoaib Akhtar     | 162.80   | 65.12%       | C
44    | Noman Ali         | 164.80   | 65.92%       | C
3     | Ahmed Raza        | 165.20   | 66.08%       | C
19    | Talha Mehmood     | 168.90   | 67.56%       | C
36    | Junaid Jamshed    | 169.50   | 67.80%       | C
48    | Kamran Akmal      | 170.20   | 68.08%       | C
23    | Fahad Mustafa     | 171.00   | 68.40%       | C
42    | Shahzaib Khan     | 173.20   | 69.28%       | C
7     | Umar Farooq       | 175.00   | 70.00%       | B
32    | Danyal Zafar      | 176.80   | 70.72%       | B
15    | Hassan Ali        | 178.40   | 71.36%       | B
```

Q: What is operating systems and how operating system do manages Process?

An **operating system** is the core software that manages a computer's hardware and software resources, acting as a bridge between users/applications and the physical machine.

**How the OS Manages Processes**

A **process** is a program in execution. The OS manages them through:

1. **Process Creation and Termination:** Starts new processes when requested and cleans them up when finished.

2. **CPU Scheduling:** Decides which process gets to use the CPU and for how long, switching between them rapidly so everything appears to run simultaneously.

3. **Process Synchronization & Communication:** Allows processes to share data safely and coordinate actions without interfering with one another.

4. **Deadlock Handling:** Prevents or resolves situations where two or more processes are stuck waiting for each other to finish.